

Abstract and Introduction	18 or greater	15 to 17	12 to 14	11 to 10	Less than 10	/20
	- The abstract gives a concise and complete summary of the project aim, the methods used, the results and their interpretation. - Background material on the topic, including a thorough survey of literature, is presented so that it is clearly understandable to a broad readership. - The relevance of the topic and the research question are clearly presented, with compelling motivation. - The contributions of the project, and an overview of the methods used, are clearly and concisely described, and compared to the literature.	- The abstract gives a concise summary of the project aim, the methods used, the results and their interpretation. - Background material on the topic, including a survey of literature, is presented so that it is understandable to a broad readership. - The relevance of the topic and the research question are presented and some motivation is provided. - The contributions of the project, and an overview of the methods used, are clearly described with reference to the literature.	- The abstract gives a summary of the project aim, the methods used, the results and their interpretation. - Some background material on the topic, including limited literature, is presented so that it is mostly understandable to a broad readership. - The relevance of the topic is briefly presented, and the research question is stated with limited motivation provided. - The contributions of the project, and a description of the methods used, are described clearly.	- The abstract gives a basic summary of the project aim, the methods used, the results and their interpretation. - Basic background material on the topic is presented, and limited engagement with literature - A basic summary of the research question is given, with limited discussion on motivation or relevance. - A basic description is given of the contributions of the project and the methods used, missing some detail.	- The abstract provides limited to no information on the project aim, the methods used, the results and their interpretation. - Limited or no background material on the topic, and no engagement with literature - The research question is not stated or easily discernible. - Limited or poor description of contributions of the project and the methods used.	
Methods	18 or greater	15 to 17	12 to 14	11 to 10	Less than 10	/20
	- Methods section provides all the detail for someone to replicate the project. - The dataset and/or model is described clearly and in detail, including key aspects of the features, sample size and sample composition. - Critical reflection provided on issues with the dataset and/or model, e.g. missing data, data preprocessing, modelling assumptions. - Technical details of the methods, including analysis	- Methods section provides most of the details to replicate the project. - The dataset and/or model is described clearly, including key aspects of the features, sample size and sample composition. - Reflection provided on issues with the dataset and/or model, e.g. missing data, data preprocessing, modelling assumptions. - Technical details of the methods, including analysis	- Methods section provides some of the details to replicate the project. - The dataset and/or model is described, along with some aspects of the features, sample size and sample composition. - Comments provided on issues with the dataset and/or model, e.g. missing data, data preprocessing, modelling assumptions. - Some discussion of the technical details of the	- Insufficient detail is provided to replicate the project. - A basic description is given of the dataset and/or model. - Limited treatment of issues with the dataset and/or model, e.g. missing data, data preprocessing, modelling assumptions. - Limited discussion of the technical details of the methods employed, including analysis	- Insufficient detail is provided to understand what has been done in the project. - Limited or no description is given of the dataset and/or model. - No treatment of issues with the dataset and/or model. - No discussion of the technical details of the methods employed.	

	techniques, key assumptions, parameter values, implementation choices, are clearly discussed. Compelling justification provided, with support from the literature.	techniques, key assumptions, parameter values, implementation choices, are discussed. Some justification provided, with support from the literature.	methods, including analysis techniques, key assumptions, parameter values, implementation choices. Limited justification and/or links to the literature.	techniques, key assumptions, parameter values, implementation choices.	
Results and Discussion	18 or greater	15 to 17	12 to 14	11 to 10	Less than 10 /20
	- Results are correctly, clearly, and concisely presented, showing critical understanding. - Comprehensive analyses, appropriately utilised to address the proposed question or problem using the methods discussed. - Results are interpreted within the context of the topic, the dataset or model, and is clearly linked to the proposed question or problem. - Consistent usage of figures to provide correct and excellent visualizations. Tables are concise and highlight important details (e.g. statistical significance). - Critical reflection provided on the limitations of the data and the methods. Suggestions for future study, including addressing the limitations, are described in detail and sufficiently motivated.	- Results are correctly and clearly presented, showing critical understanding. - The analyses are detailed, with suitable measures to address the proposed question/problem using the methods discussed. - Results are interpreted within the context of the topic, the dataset or model, and is linked to the proposed question or problem. - Consistent usage of figures to provide correct and informative visualizations. Tables are concise and highlight important details (e.g. statistical significance). - Some reflection provided on the limitations of the data and the methods. Suggestions for future study, including addressing the limitations, are described in detail.	- Results are correctly and clearly presented. - The analyses are well thought out, with some application of measures to address the proposed question or problem using the methods discussed. - Some of the results are interpreted within the context of the topic, the dataset or model, and addresses the proposed question or problem. - Some usage of figures to visualize the results. Tables highlight important details (e.g. statistical significance). - Some details provided on the limitations of the data and the methods. Suggestions for future study, including addressing the limitations, are briefly described.	- Results are presented mostly correctly. - The analyses are correct, with some application of measures to address the proposed question or problem. - Some of results are discussed or described, with basic interpretations given. The proposed question or problem has been partly addressed against the reported results. - Limited usage of figures to visualize the results. Tables contain some details. - Some mention of the limitations of the data and the methods. Suggestions for future study, including addressing the limitations, are briefly described.	- Results are presented incorrectly. - The analyses are incorrect, with no application of measures to address the proposed question or problem. - Results are not discussed or described, or in limited detail. The proposed question or problem has not been addressed. - Poor usage of figures to visualize the results. Tables do not contain important details (e.g. statistical significance). - No mention of the limitations of the data and the methods, or suggestions for future study.

Conclusions	5	4	3	2	0 to 1	/5
	- A clear and comprehensive summary of the work conducted, including methods and findings. - Critical reflection on how the project relates to existing literature, with suitable context given.	- A clear summary is given of the work carried out and the findings. - Thoughtful reflection on the project's relation to existing knowledge and literature.	- A summary is given of the work conducted and the findings. - Some basic reflection on the context and interpretation of the study.	- A basic conclusion on the findings of the work, and no evidence of critical reflection. - Limited and shallow reflection on the context and interpretation of the study.	- Limited to no attempt to summarise the work conducted, and no reflection is provided on the context and interpretation of the study.	
Organisation and structure	10	8 to 9	7 to 8	5 to 6	0 to 4	/10
	- The report is well organised, with excellent use of sections and subsections to structure the written content, with appropriate length. - There is a clear logical flow from the start to the end of the report, and the reader is easily able to follow the report and understand the overall project.	- The report is well organised, with consistent use of sections and subsections to divide the content. - There is clear logical flow from the start to the end of the report. Readers are never confused when moving to the next section.	- The report is organised, with consistent use of sections and subsections to divide the content. - There is consistent logic in the way content appears from the start to the end of the report.	- The report is organised, with some use of sections and subsections to divide the content. - There is some logical flow from the start to the end of the report. Most of the content appears in the correct order.	- The report is unorganised, with limited use of sections and subsections to divide the content. - There is no logical flow to the report. Content and sections do not always appear in the correct order.	
Writing quality and references	5	4	3	1 to 2	0	/5
	- The writing quality is high with virtually no errors, and the material is enjoyable and intellectually engaging for the reader. Shows a deep and critical understanding of all aspects of the project. - References cover a comprehensive amount of literature, and they are used to make convincing statements.	- Sentences are well-written and logically connected. There is a consistent demonstration of understanding of the methods used and results/findings. - A range of different references are used to support statements.	- Sentences are well-written. Document can be read without the writing quality being a distraction. - References are used to support most statements.	- The writing quality is not a distraction, with occasional errors. - References are used to support some statements, and appropriate citation forms utilized.	- Sentences are poorly written, with grammatical, punctuation, and spelling errors. The quality of writing is sufficiently poor that it distracts from reading the report. - Fails to correctly use references to support statements or utilize appropriate citation forms.	
Total						/80

Guidelines for writing a research report

Each team and project will have unique features. Your supervisor will take time to explain the structure that they think will work best for your specific project, but these are some general principles you should follow.

Structure and Style

The report should contain multiple sections. In the template, we have provided a common structure of sections. Your supervisor might suggest others or might suggest ways to arrange your content into this structure, and you will be judged on how well you organise and structure your report. In general, we would expect:

Abstract

This is a short summary of your work. It should communicate the main ideas and results of the project with no extraneous details. A good rule of thumb is to use 1-2 sentences for each section of your report. This means you'd have 1-2 sentences introducing the work and relevance (intro and background), 1-2 sentences on the methods used, 1-2 sentences on the main results and 1-2 sentences for the conclusion or main takeaways.

Introduction and Background

This is where you introduce and motivate the problem you are trying to solve, or question you are trying to answer. It should start general and become more specific to your research question, and you should make sure that you are very clear on WHAT you are investigating and WHY it is important. This section will set up the future sections – e.g., the methods will be judged on their explanation of HOW you answer the question(s) that are introduced here. The level of the writing in this section (what you assume of the reader) should be appropriate to a peer taking the same course. You may embed the background literature into the introduction or create a separate section, but it should be clear how your work is embedded into the wider scholarly literature. You will be expected to cite original research and have found relevant literature beyond what you have been provided to you.

Work since last submission

If your supervisor recommends it, you may include a section after the introduction outlining the additional work you have completed since submitting your progress report for part A. This can include improvements to the original analysis as well as extensions. If you have not done any other work, you should state this.

Methods

The methods section is where you describe HOW you will answer your question. It should contain enough information to reproduce the results, but it should be written for a reader who may not read your code. This means that you should refer to variables by their context and not the variable name in the data (e.g., price of product and not price_2010), and you should avoid including chunks of code; instead, you should explain what the code is doing using mathematics and words. Try to be factual, precise and succinct. If you made a mistake in doing your analysis, do not describe it here, but instead describe what you did do and motivate/defend why you chose to do it this way. Details of methods that are important but not directly relevant to the main findings (e.g., a robustness check for a methodological decision) can be placed in an appendix.

Your report must include a link to a repository holding the code you have produced for the project. This must be publicly viewable for auditing. You may include this as a url in the report or a reference.

Results

Results presented should directly relate to the main findings or research question (e.g., do not include a histogram of every variable in the dataset unless this is truly relevant). Figures can be an excellent way to communicate results; they should be styled consistently with informative captions that explain all major features of the figure (e.g., axes, colour) and succinctly interpret findings. Figures should be legible and correctly labelled, aesthetically pleasing, and express useful information that cannot be conveyed more succinctly. All figures and tables are referenced from the main text (e.g., in Figure 1 we see that ...).

Discussion/Conclusion

In general, the discussion covers four main areas. First you summarise the main results and contextualises them relative to the literature on the subject. Then you demonstrate a clear understanding of the limitations of the analysis. Finally, you present ideas to overcome limitations that are grounded in the broader literature and articulate credible ways to extend the analysis if there was more time.

References

You should include a reference section that primarily includes peer reviewed articles and books. All references should support the main arguments in your report, be correct, and be referenced appropriately in text. We suggest using a reference manager like bibtex. You may use any citation style that you are comfortable with (e.g., APA) but you must be consistent.

Formatting

We strongly advise using the provided latex template. If you choose not to, you will be expected to closely replicate the resulting style, including cover page and formatting.

Figures and tables

Figures and tables should:

- be styled consistently with informative captions.
- legible and correctly labelled.
- express useful information that cannot be conveyed more succinctly.
- aesthetically pleasing.
- referenced from the main text.

Mathematics and equations

You must explain mathematical concepts using appropriate notation. Notation should be consistent throughout your work and not simply copied from the reference book/paper. This means a symbol should consistently represent the same quantity throughout the work.

- Equations can be numbered and referenced in the main text.
- Equations should be correctly punctuated. This means they should fit into the grammar of the text.
- Mathematical symbols are all defined.
- Equations are in written using symbols, not words or acronyms.
- Code should not be used instead of mathematics.

References and bibliography

- Bibliography is well formatted.
- Full bibliographic details of each reference are provided and are consistent across references.
- Style is consistent across references.